

Phys 247: A Modern Introduction to Physics (Fall 2026)

Version: 6 Sept 2026

Lecture MWF 9:55 – 10:45 AM Chamberlin 2241

5 credits.

Instructional Mode: In-person

Canvas Course URL: <https://canvas.wisc.edu/courses/527487>

Instructors:

Professor Jan Egedal

Office: Chamberlin 3275

Office Hours: F 1:00 – 2:00 pm

Email: egedal@wisc.edu

Professor Tony Qian

Office: Chamberlin 3287

Office Hours: M 11:00 – 12:00 pm

Email: tqian23@wisc.edu

Teaching Assistants:

Joseph Hammill

Discussion sec: 301 (W 1:20 – 2:10 PM 2124 Chamberlin)

302 (H 1:20 – 2:10 PM 2124 Chamberlin)

Lab sec: 601 (M 1:20 – 4:15 3136 Chamberlin)

602 (M 4:35 – 7:30 PM 3136 Chamberlin)

Office Hours: W 4:00 – 5:00 pm (Chamberlin 2238)

Email: jhammill@wisc.edu

Daniel Heimsoth

Discussion sec: 303 (H 2:25 – 3:15 PM 2124 Chamberlin)

304 (W 2:25 – 3:15 PM 2124 Chamberlin)

Lab sec: 603 (T 1:20 – 4:15 3136 Chamberlin)

604 (T 4:35 – 7:30 PM 3136 Chamberlin)

Office Hours: H 4:00–5:00 pm (Chamberlin 3136)

Email: dheimsoth@wisc.edu

Official Course Description: Calculus-based introduction to physics intended for Physics, AMEP, and Astronomy-Physics majors. Mechanics, waves, thermodynamics and statistical mechanics, topics in modern physics; with computation. A more mathematically rigorous and in-depth introduction to physics than the other introductory physics sequences.

Requisites: MATH 222 or concurrent enrollment

Grading:

Midterm 1 (10/7 in class)	15 %
Midterm 2 (11/13 in class)	15 %
Final exam (12/18 exam room not yet posted)	25 %
Lab	15 %
Homework	20 %
Best of 3 Exams*	10 %

*the best score between two midterms and final will contribute an extra 10% of your total grade.

If you score the following percentages, you will score the letter grade or higher at the right

92 – 100 %	A
87 – 91 %	AB
82 – 86 %	B
77 – 81 %	BC
70 – 76 %	C
60 – 69 %	D
0 – 59 %	F

In other words, we reserve the right to curve the class scores, but the curve cannot lower your letter grade.

Homework Information/Policies:

Homework will be assigned weekly on Friday (or before).

They are due at the start of class on the date stated in the assignment, expected to be two Mondays later. Solutions will be released the same day. There are no late HWs.

The HW can be submitted either as handwritten paper, scanned PDF, or latex PDF.

Please keep your work neat and readable.

You are encouraged to work together on homework, but each student must submit their own assignment.

AI Expectation: You may use AI to explain the textbook and discuss unassigned problems. You may not use AI to answer, check, or polish assigned homework problems.

Lab Information/Policies:

Attendance of labs is required.

Your lab Teaching Assistant will be giving you more information about the lab during your first Discussion session.

You must attend the lab section for which you are signed up for.

There are NO scheduled labs for the first week, last week, Thanksgiving week, and the 2 midterm exam weeks.

Discussion Section:

There are weekly 50-minute discussion sessions led by your TA. This will be the same TA as your lab.

Textbook:

An Introduction to Mechanics by Daniel Kleppner and Robert Kolenkow, Second Edition

Learning Outcomes:

- Understand the relationship between forces and motion by 1) calculating the motion of an object given the forces acting on it and 2) calculating the forces acting on an object given its motion.
- Calculate the motion of objects under various constraints, such as linearly constrained motion, circularly constrained motion, and coupled objects.
- Apply conservation laws (linear momentum, angular momentum, energy) to systems of particles to understand and predict their motion.
- Describe and calculate properties of planetary orbits using Newton's laws and conservation principles.
- Translate back and forth between physical concepts and the mathematics describing the motion of objects.
- Perform measurements and analyze results in laboratory experiments using concepts from this course.

Academic Policies and Statements

Academic Integrity

By virtue of enrollment, each student agrees to uphold the high academic standards of the University of Wisconsin–Madison; academic misconduct is behavior that negatively impacts the integrity of the institution. Cheating, fabrication, plagiarism, unauthorized collaboration and helping others commit these previously listed acts are examples of misconduct which may result in disciplinary action. Examples of disciplinary sanctions include, but are not limited to, failure on the assignment/course, written reprimand, disciplinary probation, suspension or expulsion.

Accommodation for Students with Disabilities

The University of Wisconsin–Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12) and UW–Madison policy (UW-855) require the university to provide reasonable accommodations to students with disabilities to access and participate in its academic programs and educational services. Faculty and students share responsibility in the accommodation process. Students are expected to inform faculty of their need for instructional accommodations during the beginning of the semester, or as soon as possible after being approved for accommodations. Faculty will work either directly with

the student or in coordination with the McBurney Disability Resource Center to provide reasonable instructional and course-related accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA.

If you need accommodations set up, PLEASE come talk to us sooner (i.e. this week) rather than later.

Schedule

(Version 6 Sept, subject to change)

Week	Monday	Wednesday	Friday	Lab
1 Vectors Ch1		2-Sep 1.1-1.7 Vector Algebra	4-Sep 1.8-1.11 Kinematics	No lab
2 Newtons Laws Ch2	7-Sep Labor Day	9-Sep 2.1-2.5 Newtons Laws	11-Sep 2.7-2.10 Applications	No lab
3 Forces and EoM Ch3	14-Sep 3.1-3.4	16-Sep 3.1-3.4	18-Sep 3.5-3.7	M-201ac Measurements, error, and force
4 Momentum Ch4	21-Sep 4.1-4.4	23-Sep 4.5-4.7	25-Sep 4.8-4.10	M-204 Acceleration in free fall
5 Energy Ch 5	28-Sep 5.1-5.5	30-Sep 5.6-5.8	2-Oct 5.9-5.11	M-205 Projectile motion
6	5-Oct Review Day	7-Oct Midterm Exam 1 (in class)	9-Oct Ch 6 - Stability and Normal Modes	Make up
7 Angular Momentum Ch 7	12-Oct 7.1-7.4	14-Oct 7.5-7.7	16-Oct 7.8-7.9	M-211b Air track collisions
8 Rigid Body Motion Ch 8	19-Oct 8.1-8.3	21-Oct 8.5-8.7	23-Oct Ch 8 cont.	M-206 Uniform circular motion

9	26-Oct	28-Oct	30-Oct	M-203 Static Force and Moments
Non Inertial System Ch 9	Ch 9	Misc	Misc	
10	2-Nov	4-Nov	6-Nov	M-209 Angular acceleration and moments of inertia
Central Force Ch 10	10.1-10.3	10.4-10.6	11.1-11.3	
11	9-Nov	11-Nov	13-Nov	Make up
Harmonic Oscillator Ch 11	11.4-11.6	Review Day	Midterm 2 (in class)	
12	16-Nov	18-Nov	20-Nov	M-210 Young's Modulus of Elasticity and Hooke's Law
Special Relativity Ch 12	12.1-12.4	12.5-12.7	12.8-12.10	
13	23-Nov	25-Nov	27-Nov	"207" Transverse Waves on a string
Relativistic Dynamics Ch 13	13.1-13.3	13.4-13.6	Thanksgiving break	
14	30-Nov	2-Dec	4-Dec	M-212 Simple Harmonic Motion and Resonance
Spacetime Physics Ch 14	14.1-14.3	14.4-14.6	misc	
15	7-Dec	9-Dec		Make up
	misc	misc		
Finals Week				